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🌾 AgriScan: Empowering Agriculture with AI



AgriScan is a cutting-edge, AI-powered agricultural platform designed to bridge the gap between technology and traditional farming. By integrating advanced Computer Vision, Multilingual LLMs, and a decentralized Marketplace, AgriScan empowers farmers to make data-driven decisions and connect directly with the agricultural ecosystem.

🚀 Vision

To build a sustainable agricultural future where every farmer has an AI agronomist in their pocket, reducing crop loss and ensuring global food security.

✨ Key Features

🔍 AI Crop Diagnosis & Gatekeeper

- **Multi-Crop Support**: Detects diseases in Maize, Potato, Tomato, Beans, Wheat, and more.
- **Smart Filtering**: Integrated **EfficientNetB0 Gatekeeper** ensures only plant-related images are processed, increasing system trust.
- **Instant Analysis**: High-speed leaf scan analysis powered by MobileNetV2.
- **97% Accuracy**: Fine-tuned deep learning models ensuring reliable diagnosis.

🤖 AgriBot: Multilingual AI Assistant

- **RAG-Enabled**: Uses Retrieval-Augmented Generation (LangChain + ChromaDB) for context-aware advice.
- **Multilingual**: Supports **English** and **Kiswahili** to cater to diverse farming communities.
- **Tool-Calling**: Capable of fetching real-time weather data to provide location-specific advice.

🏪 Role-Based Marketplace (RBAC)

- **Farmer Profile**: List produce, find buyers, and get specialized advice.
- **Buyer/Market**: Discover fresh produce directly from the source.
- **Agro-Agencies**: Connect with farmers to provide fertilizers, seeds, and pesticides.

💰 Micro-Lending & Farm Health Score

- **Dynamic Scoring**: Generates a verifiable Farm Health Score (300-850) factoring in AI scan frequency, healthy crop ratios, and marketplace transactions.
- **Fintech Integration API**: Secure, token-authorized REST API for partnered financial institutions to retrieve real-time farmer risk tiers.
- **Consent-based Data Sharing**: Built-in privacy controls allowing farmers to explicitly opt-in to share their digital footprints to unlock non-collateral loans.

📱 Native Mobile Experience

AgriScan is optimized for mobile via **CapacitorJS**, enabling:

- **Native Camera Integration**: High-resolution image capture for AI scanning.
- **Offline Readiness**: Future support for TFLite edge inference.
- **Android/iOS Packaging**: Standardized builds for deployment on mobile app stores.

🛠️ Technical Architecture

Artificial Intelligence Stack

- **Vision Model**: `EfficientNet`/`MobileNetV2` backbone with custom transfer learning.
- **Gatekeeper OOD Filter**: `EfficientNetB0` with an ImageNet-based whitelist.
- **Embedding Engine**: `all-MiniLM-L6-v2` via HuggingFace for semantic search.
- **Vector Database**: `ChromaDB` for high-performance knowledge retrieval.
- **LLM Reasoning**: Powered by **Groq / Llama-3.3-70b** for ultra-fast, intelligent conversations.

Backend Infrastructure

- **Framework**: Django 5.x with Django REST Framework.
- **Storage**: **Cloudinary** for persistent media storage in production.
- **Database**: PostgreSQL (Production) / SQLite (Development).
- **Security**: Role-Based Access Control (RBAC), Rate Limiting (`django-ratelimit`), and CSRF Protection.

📦 Installation & Setup

Local Development

1. **Clone the Repository**:

```
git clone https://github.com/jobcode003/AgriScan.git
cd crop_AI_analyzer
```

2. **Setup Virtual Environment**:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

3. **Configure Environment Variables**: Create a `.env` file and add:

```
SECRET_KEY=your_key
DEBUG=True
GROQ_API_KEY=your_groq_key
HF_TOKEN=your_huggingface_token
CLOUDINARY_URL=cloudinary://api_key:api_secret@cloud_name
```

4. **Run Migrations & Start Server**:

```
python manage.py migrate
python manage.py runserver
```

Production Deployment (Railway)

AgriScan is pre-configured for **Railway** via Docker.

> [!IMPORTANT] Ensure the `CLOUDINARY_URL` is set in your Railway environment variables to enable persistent storage for user uploads.

📊 Analytics

Metric	Performance
Training Accuracy	97.2%
Validation Accuracy	85.6%
Average Inference Time	240ms
Model Type	MobileNetV2 (Fine-tuned)

🤝 Contributing

Contributions are what make the open-source community such an amazing place to learn, inspire, and create. Any contributions you make are **greatly appreciated**.

1. Fork the Project
2. Create your Feature Branch (`git checkout -b feature/AmazingFeature`)
3. Commit your Changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the Branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

📄 License

Distributed under the MIT License. See `LICENSE` for more information.

📬 Contact

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Project Link: https://github.com/jobcode003/AgriScan